LIBBY — PATIENT/CAREGIVER TRACK

gist-sdh-multifocal-resected-m1-k4n8

Plain-language summary for patient and caregiver

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What this page is

Libby is a decision-support tool. It read your case, looked at the published research on GIST and at what's available right now (including clinical trials), and put together a ranked list of what to do next. This is the patient and family version. The clinician's version, with the technical details, lives on a separate page.

Read this as a starting point for a conversation with your oncologist, not a treatment plan. A few real uncertainties are baked in. The genetics aren't fully nailed down yet. You didn't fill in a preferences form. And the kind of GIST you have is rare enough that the published evidence is thinner than anyone would like. All of that is called out below.

What we know about your cancer

You have a kind of stomach cancer called a gastrointestinal stromal tumor, or GIST. Most GISTs are driven by mutations in one of two genes, KIT or PDGFRA. Yours isn't. Your tumor is what's called **SDH-deficient GIST** (sometimes shortened to dSDH-GIST). It's a rare subtype, behaves differently from the common kind of GIST, and importantly does not respond well to the drug most people with GIST hear about first (imatinib, brand name Gleevec). More on that further down.

The other facts that shape this page:

- You had **multifocal disease**: more than one tumor, all in the stomach, plus one spot of spread to a structure near the small intestine called the ligament of Treitz. That counts as M1 (metastatic) by staging rules, even though it was a single, surgically removable spot.
- You had a **distal gastrectomy with a Roux-en-Y reconstruction** (a partial stomach removal with the small intestine reconnected in a Y-shape). The surgeon got everything: clean margins, no cancer in the seven lymph nodes that were checked, and the spot near the small intestine was completely removed.
- You are currently **NED, no evidence of disease**. That's the best possible starting point.
- Your **performance status is good** (ECOG 1, meaning you're up and around with maybe some limits on strenuous activity).
- Your tumor's **molecular profile is mostly worked out, but not fully locked in.** Two mutations in a gene called SDHA were found in the tumor. A few other possibly-relevant mutations showed up in a blood-based (ctDNA) test but haven't been confirmed in tumor tissue yet. And there's a technical puzzle the pathology lab needs to resolve: an immunostain called SDHB came back "intact" when, given the SDHA findings, the textbook says it should have been lost. That paradox is real and needs sorting out before some downstream decisions can be made. The first item below is the workup that does exactly that.

What you told us matters most

You didn't actually tell us anything explicit. **The preferences on file are defaults**, picked because they seemed reasonable for a person in your situation:

- A mild lean toward effectiveness over avoiding side effects, with the caveat that you don't have measurable cancer right now, so the bar for committing to long-term drug side effects is high.
- An openness to clinical trials, partly because the most targeted options for SDH-deficient GIST sit mostly in trial form.

- No specific things you said you wanted to avoid.
- Your geography (where you live, how far you can travel for care) is unknown.

This matters. A few of the contingency plans below depend on where you live and how willing you are to travel. One trial is in Korea. One is at a single site in Boston. A few are scattered across about a dozen US cities. The right way to use this page is to actually elicit your preferences (and your geography) with your oncologist or a genetic counselor before any of the contingency picks gets committed to.

The first step everyone agreed on

Before any of the contingency drug picks below can move forward, the genetics need to be locked in. This is the first thing to do, and it has no real downside.

Three answers come from one tissue pull and one blood draw:

1. **Is the SDHA mutation in your blood, or only in your tumor?** This is the germline-versus-somatic question. A blood test (or saliva) checks the SDHA gene from your normal cells. If the mutation is also there, you're a carrier. That has implications for your relatives (they may want to be tested too) and for your own lifetime monitoring for related conditions (see Option 3). If the mutation is only in the tumor, it's a tumor-only event and the family conversation looks different.

1. **What does the SDHA immunostain look like, and what happens when the SDHB stain is repeated?** This is the paradox mentioned above. The pathology lab re-cuts and re-stains the existing tumor block. A specialized reference lab is the safest place to do it; Mayo Clinic Laboratories is the usual pick because they have a validated protocol. The point is to either resolve the paradox (the SDHB stain comes back as lost when done carefully) or to deepen it (it really is intact, in which case the diagnosis gets re-examined).

1. **Are the three mutations seen only in blood (PIK3CA R93W, MAP2K1 P124S, and a third SDHA hit called E350fs) actually in the tumor?** A repeat panel on the tumor tissue answers this. If they are, the picture sharpens. If they aren't, those mutations drop out of the conversation entirely.

All three answers come from the same stored tumor block plus one venipuncture. No drugs, no side effects. The cost is **about two to four weeks of waiting** for results, which is acceptable: you have no measurable cancer right now and the answers shape every downstream decision.

Here's the reason this comes first. Every clinical trial for SDH-deficient GIST uses one of the SDHB or SDHA stains as the entry ticket. If the paradox is left unresolved, you could turn up at a trial screening and get bounced. Lock the diagnosis in now so that doesn't happen.

While you're at it, the same blood draw can include a **screen for paraganglioma and pheochromocytoma** (Option 3 below). Operationally efficient, separate result paths, one needle stick.

The options

Twelve options were considered. Some are things to do right now (surveillance, registry enrollment, a few baseline screens). Some are contingencies, meaning what to reach for if and when the cancer comes back. One is on the page specifically to say "no" to it (adjuvant imatinib).

The options are presented in rank order, but rank is not a strict sequence: the first several should all happen together as part of an active-but-non-toxic management plan. The contingency drugs are stacked for if recurrence happens; you would not be on more than one at a time.

Option 1 — Active surveillance, no adjuvant drug treatment

This is the lead option for what to do right now. It means regular scans every 3 to 6 months for the first 2 to 3 years, spaced out after that. No drug treatment in the meantime.

Why this is the call: the standard "first-line GIST drug" everyone knows about is imatinib (Gleevec). It's a real success story in the most common kind of GIST. In your kind, SDH-deficient GIST, it doesn't work. In a published case series of patients with SDH-deficient GIST given imatinib, only 1 out of 49 people had a partial tumor response. That's a 2% response rate. The side-effect burden of three years on imatinib includes meaningful rates of swelling and fatigue plus ongoing liver and heart monitoring. Trading three years of side effects for a 2% chance of benefit is not a deal worth taking.

Both of the major cancer society guidelines (NCCN in the US, ESMO in Europe) explicitly recommend against adjuvant imatinib for SDH-deficient GIST. This is one of the clearest decisions in the entire case.

About the recurrence math, since you'll want to know: based on a 76-patient cohort followed at the NIH, the median time before recurrence after complete surgery for SDH-deficient GIST is roughly 2.5 years, and around 71% of people eventually have a recurrence. That's high. The flip side is that the disease tends to grow slowly when it does come back, so scans pick it up at a point where there are still options. The surveillance approach isn't passive. It's the active choice to keep every downstream option open instead of burning months or years of toxicity now on a regimen the evidence rejects.

The honest downsides:

- **Scan anxiety is real.** Indefinite follow-up imaging doesn't show up on any side-effect chart, but it shows up in real life. Raise it with your care team if it gets heavy.
- **Your case is on the harder end of the natural-history range.** The 2.5-year median time-to-recurrence number comes mostly from people with a single localized tumor. You had multifocal disease plus a single spot of spread. That probably puts you at the higher-risk end of the cohort, so the tighter end of the cadence (closer to every 3 months than every 6) is the right starting point.
- **"Do nothing" isn't quite the right framing.** The cadence, what gets imaged, the conversation about your relatives, the lifetime monitoring for related conditions: that is the active part. Surveillance plus the next several options is the plan.

Option 2 — Enroll in the NCI Rare Tumor Natural History Study and see a cancer-genetics counselor

This is a research registry at the NIH, with alternative sites at OHSU in Oregon and Texas Children's. It's open to people with SDH-deficient GIST regardless of whether they have measurable disease, which means **you can enroll now, in your NED state, while nothing else can be enrolled in.**

What you get:

1. **Longitudinal specialty follow-up at the NIH Pediatric and wild-type GIST Clinic.** This is the reference

center in the US for your subtype. SDH-deficient GIST is rare enough that most community practices won't see many cases. Staying connected to a center that does is real value.
2. **Banked tissue and blood samples** that pay forward if a trial slot opens later. When a sponsor opens enrollment for an SDH-deficient GIST trial, having your samples already in the system is the operational difference between making the screening window and missing it.
3. **Access to a specialized test called SDHC promoter methylation**, which is almost impossible to get outside of an academic referral pathway. If the workup above takes an unexpected turn, this test becomes important. The registry is the door to it.

Pair the registry enrollment with a visit to a **cancer-genetics counselor**, ideally after the germline SDHA blood result comes back. Two things to discuss:

- The SDHA germline result, whatever it is, and what it means for your siblings, parents, and children. If you are a carrier, first-degree relatives can be tested. The decision about who to tell and when is yours; a counselor walks you through it.
- A separate finding the tumor panel turned up: a variant in a gene called POLE, specifically R1679C in your germline. In some cancers POLE variants are very actionable. In your case, the variant sits outside the part of the gene that drives that biology, and your tumor's other markers (TMB <1, MSS) point the other way. The most likely landing place is "variant of uncertain significance," meaning a known gene change that doesn't currently change anything you do. Expect that outcome going in, so the result doesn't get over-interpreted when it lands.

Cost: one in-person visit at NIH Bethesda (or OHSU, or Texas Children's). Travel is the main operational expense. **Side effects:** none.

Option 3 — Baseline screen for paraganglioma and pheochromocytoma

If the SDHA germline test confirms you are a carrier, you have a lifetime increased risk for two rare tumors that come from the adrenal glands and related tissue: pheochromocytoma and paraganglioma (often shortened to PCC and PGL). They're usually treatable when caught. But they have one feature that matters urgently. An unrecognized, hormonally-active one can cause a dangerous blood-pressure spike during anesthesia. If you ever need another surgery (possible if cancer comes back), the surgical team needs to know whether you have one.

The baseline workup:

- A blood test for plasma free metanephrines (the standard biochemical screen).
- A whole-body MRI, skull-base to pelvis, as the anatomic backbone. MRI is preferred because it doesn't add radiation, so it can be repeated long-term.
- A specialized PET scan called Ga-68 DOTATATE is the most sensitive test, but it does add a small radiation dose. The right use is one baseline plus a repeat only if something changes, not as part of the routine cadence.

If the germline SDHA test is negative (no carrier status), the long-term cadence relaxes. If it's positive, the imaging becomes a roughly every-other-year commitment for life. Either way, the baseline is worth doing now: you want a clean comparison point.

Side effects: none. **Cost:** insurance approval for the whole-body MRI is sometimes a fight, but it's the right test.

Option 4 — Signatera ctDNA monitoring as an imaging adjunct (with one important guardrail)

Signatera is a blood-based test that looks for tiny amounts of tumor DNA in your bloodstream. The test is customized to your specific tumor's mutations, which is convenient here, because your tumor's mutations showed up clearly in blood before surgery. A Signatera panel built from your case should be a reliable signal.

What it gives you is an **early-warning signal**. In other cancers (colorectal and lung), a positive Signatera result tends to show up weeks or months before a scan would catch the same recurrence. In SDH-deficient GIST specifically, the evidence is thinner. This is borrowed-strength reasoning, not studied-in-your-cancer reasoning.

The guardrail is important. **A positive Signatera result does NOT mean it's time to start drug treatment.** It means it's time to get a closer-spaced scan, around six to eight weeks out instead of waiting for the next three-to-six-month slot. **Only an imaging-confirmed recurrence triggers the contingency plan below.** Why so strict: blood-based MRD testing in GIST hasn't been validated to the point where a ctDNA-positive, scan-negative result should change drug treatment. Without that guardrail in writing, this assay becomes an anxiety amplifier with a premature-treatment risk attached.

Talk to your oncologist about putting that guardrail in the chart note before the first draw.

Side effects: none (a blood draw). **Cost:** insurance-dependent.

Options 5–10 — Contingency plans for if the cancer comes back

These options are not on the table today. You have no measurable disease, and starting drug treatment now would burn options and add side effects for no good reason. They are on the page so that if and when a recurrence is detected, the conversation can move fast and the decision is informed.

If recurrence happens, **two parallel conversations should run side by side**: the standard targeted drug (Option 5, sunitinib) and the clinical-trial track (Options 6, 7, 8). Which is the right pick depends on the genetics workup (especially whether the SDH-deficient diagnosis is fully confirmed), where you live, and what you actually want. None of them is a clear winner on the published evidence.

Option 5 — Sunitinib (Sutent), when there is measurable cancer to treat

If your cancer comes back in a measurable form (visible on a scan), **sunitinib is the best-characterized standard drug for SDH-deficient GIST in this situation.** It's FDA-approved for GIST after imatinib fails. In the published research, the response rate in SDH-deficient GIST specifically is about 18% (7 out of 38 patients in a published series). That's not a high number, but it's much better than imatinib's 2% in the same subtype, and the published evidence is the most solid of any drug option on this page.

Side effects to know about: hand-foot skin reaction (painful redness and peeling on palms and soles), low thyroid function (your TSH gets checked every cycle), fatigue, high blood pressure, and a skin color change. These are well-known, and oncologists have detailed playbooks for managing each one. Most people with SDH-deficient GIST end up needing a dose reduction within the first two or three cycles, from the labeled "50 mg four-weeks-on, two-weeks-off" schedule to a lower continuous dose. Plan for that as the normal trajectory, not a failure.

One sequencing point: because you'd be open to a trial, the trial options below should be raised as a side-by-side conversation at the moment of recurrence, not treated as something to try only after sunitinib. They may have higher upside; the trade-off is that the trial evidence is thinner and the side-effect profiles are different.

Option 6 — Belzutifan clinical trial (LITESPARK-015), gated on diagnosis confirmation

This option is only available if the SDH-deficient diagnosis comes back confirmed. If the workup at the top of the page resolves the paradox in the direction of "yes, this really is SDH-deficient GIST" (the SDHB stain comes back as lost, or the SDHA stain confirms the loss), then this trial is open. If the workup goes the other way, with both stains coming back intact and the diagnosis getting re-examined, this trial is foreclosed entirely. The trial uses the stain result as the entry ticket.

What it is: belzutifan is a drug that blocks a protein called HIF-2 alpha, which is part of the biological chain that gets activated in SDH-deficient cancers. The strongest evidence so far comes from a related rare tumor (paraganglioma), where belzutifan produced a response rate of about 26%. The trial for SDH-deficient GIST is open and recruiting at 14 US sites, so geography is more manageable than the next two options. The GIST-specific results haven't been published yet; the readout is expected sometime in 2026.

Side effects to know: belzutifan suppresses red blood cell production, which causes anemia and shortness of breath in roughly 25 to 30% of patients. **Your Roux-en-Y surgery complicates this.** That reconstruction reduces how well you absorb iron and vitamin B12, and adding a drug that lowers red blood cells on top of that is a stacked problem. Your iron, B12, and hemoglobin numbers need to be documented at baseline, and a plan for managing them needs to be in place before enrollment.

Option 7 — Temozolomide clinical trial (or off-label temozolomide), gated on diagnosis confirmation

Same gate as Option 6: only on the table if the SDH-deficient diagnosis is confirmed.

What it is: temozolomide is an oral chemotherapy that has been used for decades in brain tumors. There's a small but real signal that it works in SDH-deficient GIST. In a published five-patient cohort, two patients had a partial response. That's 40% on the surface, but with only five people, the true range could land anywhere from 5% to 85%. The number is suggestive, not reliable. The harder evidence is mechanistic: temozolomide damages tumor DNA in a way that SDH-deficient cells seem unusually vulnerable to.

The active trial is at Asan Medical Center in Seoul, South Korea. That is a hard geography gate. There's a second trial at UCSD that closed enrollment and is expected to publish results soon. That readout (due to complete in mid-2025) will tell us whether the small case-series signal holds up at a bigger size. There's also an off-label US route. Your oncologist can prescribe generic temozolomide, and a test on your tumor block called **MGMT methylation** can strengthen the case to your insurance for covering it.

Side effects: temozolomide is well-known. The main near-term issues are low blood counts, nausea, and fatigue. The longer-term concern is more serious. With cumulative exposure beyond 6 to 12 months, there's a roughly 1% risk of a treatment-related blood cancer (MDS or AML). For someone who might be on the drug for years if it works, that risk compounds. Build a scheduled re-evaluation point at 9 to 12 months into the plan.

Option 8 — Pemigatinib clinical trial (PEMIGIST at Dana-Farber), gated on diagnosis confirmation

This option carries caveats. Same diagnosis gate as Options 6 and 7. And there are honest concerns about the strength of the evidence and the side-effect monitoring burden, covered below.

What it is: pemigatinib is a drug that blocks a family of growth-signaling proteins called FGFRs. In an SDH-deficient GIST tumor model in mice, pemigatinib shrank tumors that didn't respond to sunitinib or regorafenib. A related drug in the same family was tested in a 24-patient cohort and produced a response rate of about 42%. With only 24 patients the true range is wide (roughly 22% to 63%), so the answer could be modestly active or very active, and 24 isn't enough to tell.

The trial is at Dana-Farber Cancer Institute in Boston, single-site, run by Dr. Suzanne George. That's a hard geography gate unless you're in Boston or willing to travel there.

The caveats:

- **The evidence base is shallow.** One mouse-model study (in a conference abstract, not a full peer-reviewed paper), plus a 24-patient cohort of a related drug. There's a real question about whether that's enough to pin a recommendation on.
- **The side-effect monitoring is intensive.** About 25% of patients on FGFR-class drugs develop a retinal detachment (a separation of the back of the eye), so eye exams (an OCT scan) are needed at baseline and monthly through the first 6 months. Roughly 60% have elevated phosphate levels in their blood, which needs weekly monitoring during dose adjustment and often a phosphate-binding medication. Nails and the lining of the mouth also commonly take a hit.

This is on the page because the mechanism story is the freshest in SDH-deficient GIST and the early signal is interesting. It's at this rank, with caveats, because the published evidence is thinner than the other two trial options and the monitoring burden is real.

Option 9 — Regorafenib, third-line standard drug

If you recur, take sunitinib (Option 5), and the sunitinib eventually stops working, regorafenib is the next standard-of-care drug. **This option carries caveats**, the same concern that surfaced for sunitinib: a fresh clinical-trial review should run alongside before defaulting to regorafenib at that moment, because the SDH-deficient GIST trial landscape may well have moved by the time you reach this point.

Regorafenib is FDA-approved for GIST and has solid evidence behind it. As with sunitinib, the evidence comes from a population that was mostly the more common KIT-mutant GIST, not your subtype.

Side effects are similar in flavor to sunitinib's, with higher rates of severe high blood pressure (about 23%) and hand-foot skin reactions (about 20%). Dose modifications are essentially the rule, not the exception.

Option 10 — Olaparib plus temozolomide (combination drug), late-line and bounded

This option carries strong caveats. It's on the page because of an interesting biology argument. SDH-deficient tumors have a DNA-repair weakness that should make them sensitive to a class of drugs called PARP inhibitors (olaparib is one). Paired with temozolomide, the combination should hit the cells harder than either alone.

The clinical evidence: a single published case report. One patient with heavily pre-treated SDH-deficient GIST and a related tumor had a durable response on the combination. n=1.

The concerns are serious:

- **Stacked blood-cancer risk.** Both olaparib and temozolomide carry a roughly 1% long-term risk of treatment-related blood cancers (MDS or AML). Together, in someone who stays on the combination for a sustained response, that risk compounds in a way that hasn't been studied.
- **Off-label and expensive.** Olaparib for this use isn't insurance-covered by default and runs roughly $17,000 per month if you pay out of pocket.

This option is on the page **only in a tightly bounded framing**: very late line (after the standard drugs and the trial options have been exhausted), only through a clinical trial or compassionate-use pathway, with a blood specialist co-managing, with a baseline bone-marrow assessment, and with a pre-set re-evaluation point at 9 to 12 months.

As an off-label, out-of-pocket option without that structure, it should not be done. That's the most cautious read of the case, and the right one.

Option 11 — Adjuvant imatinib (Gleevec)

This option was considered but not recommended. It's on the page specifically to make the "no" explicit, because imatinib is the famous GIST drug, and the reflex from any team that hasn't worked with SDH-deficient GIST is to reach for it.

The reasoning is the same as in Option 1 above. In published case series, the response rate of SDH-deficient GIST to imatinib is about 2%. The side-effect burden of three years on the drug is meaningful. Both NCCN and ESMO guidelines explicitly recommend against using imatinib as adjuvant therapy in SDH-deficient GIST. The rejection was unanimous.

What to ask for: a clear note in your post-operative chart that adjuvant imatinib is not recommended for SDH-deficient GIST, citing the ESMO guideline and the published case series. That way the decision is on record and doesn't get re-litigated by a community oncologist later.

What happens if the diagnosis doesn't confirm

Real possibility: the workup at the top of this page is done, and the SDH-deficient diagnosis doesn't lock in cleanly. The SDHB stain still reads as intact after a careful re-do, the SDHA stain also reads as intact, and the genetics don't produce a clear constitutional explanation. In that case, **the trial options at Options 6, 7, 8, and 10 are off the table.** The SDH-deficient diagnosis is the entry ticket, and without it, those trials can't enroll you.

If that happens, **the recommendations on this page are scoped to SDH-deficient GIST. That's the targetable feature this page is built around, and there are no other ranked options here that would replace those trial picks.** Standard care for second-line GIST that isn't SDH-deficient and isn't KIT-mutant exists, and there's a backup specialized test (SDHC promoter methylation) that the NIH registry can arrange. But those are conversations with your oncologist and the NIH clinic, not options ranked on this page. If the diagnosis falls apart, this page becomes much shorter, and the path forward is a separate discussion through your treating team.

Questions to ask your oncologist

1. **The workup.** Can we send the tumor block to Mayo Clinic Laboratories for the SDHB re-cut plus an SDHA stain, run a comprehensive tissue NGS panel (FoundationOne CDx or equivalent) on the same block, and order the germline SDHA panel on blood (Labcorp Genetics Hereditary Paraganglioma-Pheochromocytoma Panel)? Can we do all three together so we get the answers in one pull?

1. **Surveillance cadence.** Given the multifocal disease and the M1 deposit, can we start on the tight end of the schedule (every 3 months for the first year or two) rather than the every-6-month end? And can the imaging modality be MRI-based where possible to limit radiation?

1. **The chart note on imatinib.** Can you document in my post-op note that adjuvant imatinib is not recommended for SDH-deficient GIST, citing the ESMO 2022 guideline and the Boikos 2016 paper, so this doesn't get re-litigated?

1. **The Signatera guardrail.** If we use Signatera ctDNA monitoring, can you put in the chart that a positive result

triggers shorter-interval imaging (6–8 weeks), not a drug-treatment decision?

1. **Family.** Once the germline SDHA result is back, what do I tell my parents, siblings, and any children, and is a genetics counselor available to help with that conversation?

1. **The pheochromocytoma/paraganglioma screen.** Can we do the baseline plasma metanephrines and whole-body MRI now, before I might need any future surgery, so we have a clean baseline for anesthesia safety?

1. **The registry.** Can you refer me to the NCI Rare Tumor Natural History Study at NIH (or OHSU, or Texas Children's)? Is there a preferred timing relative to the germline SDHA result?

1. **If the diagnosis doesn't confirm.** If the SDHB and SDHA stains both come back intact after a careful re-do, what's our next step? Is SDHC promoter methylation testing on the table, and would I need a referral to NIH to arrange it?

1. **At recurrence (looking ahead).** If and when something shows up on a scan, can we plan for a side-by-side review of the trial-route options (belzutifan, temozolomide, pemigatinib) alongside sunitinib at that moment, rather than defaulting to sunitinib first?

Sources

The recommendations on this page rest on the following published studies (PMIDs are PubMed identifiers; you can look them up at pubmed.ncbi.nlm.nih.gov) and clinical-trial registrations (NCT numbers are at clinicaltrials.gov):

Key references:

- PMID 27011036 — Boikos 2016, the NIH series showing the 2% imatinib response rate and the 18% sunitinib response rate in SDH-deficient GIST.
- PMID 29413424 — Mei 2018, the natural-history review of SDH-deficient GIST after surgery.
- PMID 17046465 — Demetri 2006, the pivotal trial for sunitinib in GIST.
- PMID 23177515 — Demetri 2013 (GRID), the pivotal trial for regorafenib in GIST.
- PMID 34426440 — Yebra 2022, the temozolomide case series and tumor-model work.
- PMID 39927693 — Florou 2025, the contemporary review of how to work up an SDH-deficient GIST.
- PMID 35546442, PMID 23023976, PMID 23046294, PMID 23459398 — the technical papers on the SDHB/SDHA immunostain pitfall.
- PMID 31369093, PMID 23934599, PMID 24893135 — the Endocrine Society guidance on baseline screening for paraganglioma and pheochromocytoma.
- PMID 34754095 — the ctDNA monitoring (MRD) reference for the Signatera approach.
- PMID 36151992 — Singh 2023, the single-patient case report on olaparib plus temozolomide.
- PMID 30013182, PMID 32494005 — Sulkowski 2018 and 2020, the mechanism papers on the DNA-repair weakness in SDH-deficient cells.

Open clinical trials referenced:

- NCT03739827 — NCI Rare Tumor Natural History Study.
- NCT04924075 — LITESPARK-015 (belzutifan).
- NCT05661643 — Temozolomide trial at Asan Medical Center, Seoul.
- NCT03556384 — Temozolomide trial at UCSD (results pending).
- NCT07434843 — PEMIGIST (pemigatinib at Dana-Farber).